Republic of the Philippines
PARTIDO STATE UNIVERSITY
College of Engineering and Computational Sciences
Department of Computational Sciences
Goa, Camarines Sur

AURORA: ACADEMIC UNIFIED REVIEW, OBSERVATION, RATING, AND ASSESSMENT — A PAPERLESS ACADEMIC DEFENSE WORKFLOW AND QUALITY ASSURANCE PLATFORM

A Capstone Project Manuscript
Presented to the Faculty of the Department of Computational Sciences
College of Engineering and Computational Sciences
Partido State University

In Partial Fulfillment
of the Requirements for the Degree
Bachelor of Science in Information Technology

by:

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October 2026

APPROVAL SHEET

This capstone project entitled "AURORA: ACADEMIC UNIFIED REVIEW, OBSERVATION, RATING, AND ASSESSMENT — A PAPERLESS ACADEMIC DEFENSE WORKFLOW AND QUALITY ASSURANCE PLATFORM", prepared and submitted by GENE VINCENT MAÑAGO, CAROL ANN CASTILLO, RONA MAE BUENAFE, and LARA MAE ENCINAS, in partial fulfillment of the requirements for the degree of BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY, has been examined and is hereby recommended for acceptance and approval for Oral Examination.

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APPROVED by the Capstone Project Oral Examination Committee on October 2, 2026 with a rating of PASSED (94.50%).

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ACCEPTED AND APPROVED in partial fulfillment of the requirements for the degree of BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY.

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ABSTRACT

Title: AURORA: ACADEMIC UNIFIED REVIEW, OBSERVATION, RATING, AND ASSESSMENT — A PAPERLESS ACADEMIC DEFENSE WORKFLOW AND QUALITY ASSURANCE PLATFORM
Proponents: Gene Vincent Mañago, Carol Ann Castillo, Rona Mae Buenafe, Lara Mae Encinas
Degree: Bachelor of Science in Information Technology (BSIT)
Institution: Partido State University — College of Engineering and Computational Sciences
Research Adviser: Prof. Froiland B. Peñalosa
Academic Year: 2025–2026 to 1st Semester, 2026–2027 (October 2026)

Undergraduate capstone defense management at Partido State University has traditionally relied on paper-intensive manuscript submissions, manual ink annotations, ad-hoc room and panelist scheduling, and hand-computed evaluation sheets. These manual workflows result in prolonged review turnaround times (averaging 18.4 days), scheduling conflicts, heavy printing costs for students, and the absence of cryptographic auditability.

To resolve these bottlenecks, this study designed, developed, and evaluated AURORA (Academic Unified Review, Observation, Rating, and Assessment), a cloud-native, paperless academic defense management and quality assurance platform for the College of Engineering and Computational Sciences (CECS). Engineered under the Agile Scrum methodology using Next.js 16 (App Router), React 19, TypeScript, and Supabase (PostgreSQL 15 with Row-Level Security), AURORA delivers four core technical contributions: (1) automated multi-stage defense workflow management with FIPS 180-4 SHA-256 manuscript version control and Form DCS-CF-03 adviser endorsement gating; (2) a dual-pane split-screen PDF.js review workspace featuring resolution-independent coordinate annotations synchronized via real-time WebSockets alongside an automated defense scheduler with room-clash and adviser Conflict-of-Interest guards; (3) a dynamic weighted rubric scoring engine (Form DCS-CF-04) secured by Republic Act No. 8792 step-up password re-authentication and sequential clearance serials (AURORA-YYYY-NNNNNN); and (4) a 3-2-1 Disaster Recovery and Self-Healing Maintenance Console.

System acceptability and usability were empirically evaluated by N = 27 ParSU CECS stakeholders (15 BSIT senior students, 5 research advisers, 5 defense panelists, 1 capstone coordinator, and 1 college dean) using an expert-validated questionnaire (Good & Scates mean = 4.89/5.00; Cronbach's $\alpha = 0.946$) combining the 10-item System Usability

Scale (SUS) and 27-item Unified Theory of Acceptance and Use of Technology (UTAUT) model. AURORA achieved a Composite SUS Score of 91.8 / 100 (Grade A — Excellent) and a Composite UTAUT Grand Mean of 4.86 / 5.00 (Highly Acceptable), while reducing manuscript endorsement turnaround time by 85.9% (from 18.4 days to 2.6 days).

Keywords: *Paperless Academic Workflow, Capstone Defense Management, Coordinate PDF Annotation, Republic Act No. 8792 Digital Signatures, Row-Level Security, System Usability Scale (SUS), UTAUT.*